30 Problem Statements for 'Agri Hackathon 2025'

Crop improvement and Biotechnology:

1. Strategies to prevent hard seed coat formation in black gram and green gram.
2. Technological interventions to arrest pre-harvest sprouting in rice and ground nut.
3. Assessing rice grain quality/chalkiness enabling fair pricing.
4. Innovative approaches to reduce shattering loss in pulses and sesame.
5. Bio-colorants from fruits and vegetables for use in food industry.

Crop and Natural Resource Management:

6. Organic way of weed management for pulse crops.
7. Technological solutions enabling rapid increase in soil organic carbon content.
8. Developing easy-to-use, rapid soil nutrient testing kits.
9. SmartBioM: Precision biomolecule delivery for crop health and defence.
10. Assessing greenhouse gas emissions and exploring carbon markets.

Horticulture:

11. Natural preservatives from *Aloe vera*, chitosan, and plant based essential oils in preserving the freshness of flowers (Jasmine) and fruits.
12. Utilizing *Lantana camara* stems as an alternative to wood in furniture or handicraft production.
13. Multi-micronutrients formulations for pandal vegetables.
14. Formulation of a universal nutrient solution for growing vegetable crops in hydroponics.

Crop protection:

15. Formulation, development and validation of eco-friendly repellents for mitigating wild boar/elephants in agricultural fields.
16. Sustainable strategies to combat rugose spiralling white fly in coconut.
17. Ways to increase bee activity.
18. Quick detection of pesticide residues through nano-metal sensors/enzyme or bioassay/AI based methodologies in fruits, vegetables, spices and plantation crops.
19. Early detection diagnostics for coconut root wilt.

Agricultural Engineering:

20. Affordable local storage innovations for storage of perishable fruits to benefit small-scale farmers.
21. Low draft power solutions suitable for small farm holdings.
22. Drone/AI based crop health monitoring system.
23. Extending shelf-life of mango pulp beyond one year.
24. Millet decorticator.
25. Harvester cum thrasher for sesame, pulses, ground nut and sugarcane.
26. Value addition in tomato.

Agricultural Marketing/Extension:

27. Promoting digital marketing tools/Apps.
28. Supply chain management for major agricultural and horticultural crops.
29. Price forecasting App development.
30. Integrating information dissemination system (weather, etc.).